

Rough Draft

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Abstract:

This study analyzes the growth rates of fossilized clams collected from lakes and rivers across Northern New York to draw conclusions about environmental conditions and how they may impact a bivalve's growth. Dr. Judith Nagel-Myers (Department of Biology and Geology) measured each clam grew during each annual period using shell growth rings. Since clam growth decreases with age, a Von Bertanfly growth model is applied to describe the relationship between shell length and age. To estimate each of the model's parameters, a Markov Chain Monte Carlo (MCMC) Algorithm was built using a Bayesian framework, which allows for uncertainty in the growth model parameters. Our objective is to determine if clam growth rate is different between lake environments and river environments.

Introduction to Data:

Clams grow through a process called accretion, where every growing period their shells expand by adding new material to the outer edge. This creates a new visible ring around the shell, similar to tree rings. The number and thickness of these rings depend on a variety of factors, such as climate, food availability, and water conditions. These rings are preserved in fossilized clam shells, allowing geologists and paleontologists to make conclusions regarding historic clams ages and environmental conditions.

The data used in this project was collected and recorded by Dr. Judith Nagel-Myers of the Geology and Biology Departments at St. Lawrence University in Canton, New York. She collected 151 fossilized valves from 17 different localities in Northern New York and Adirondack area.

The data set contains the following useful variables:

Variable	Explanation
Location	The lake or river that the clam originated from.
Body	Binary variable that indicates lake or river.
Growth Period	Indicates stage of growth. 1 growth period represents 1 year of life.
Length	Overall length of valve.
Growth	Total growth during specified growth period.

Introduction to Growth Model:

For this study, the Von Bertanfly Growth Model was chosen because it is specifically designed to describe biological growth that slows over time, a growth pattern characteristic of bivalves. Unlike linear or exponential models, the Von Bertanfly Growth Model reflects the biological reality that organisms grow rapidly in the early life stages and slow over time. This growth pattern makes the model well-suited for analyzing clam growth, especially when working with fossilized specimens where size and age relationships can be reconstructed.

The model can be expressed as:

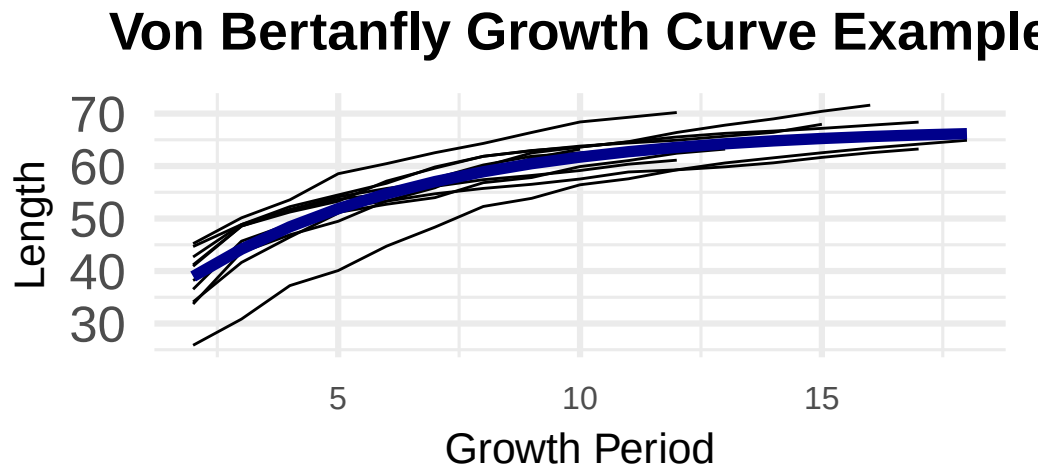
$$L(t) = max_l \cdot (1 - e^{-k \cdot (t-b)})$$

where

- t is time.
- $L(t)$ is mean length at time t .
- max_l is the asymptotic length.
- k is how quickly the clams reaches max_l .
- b is time where length is 0.

Together, these parameters allow for a detailed interpretation of growth patterns, making it possible to compare how clams develop under different environmental conditions and to infer factors such as habitat quality and resource availability from fossil data. Below is a sample of

the Von Bertanfly growth model. Take note of the exponential term which leads to the shape of the curve.



The line represents the general Von Bertanfly growth model for a clam in a specific location. Black lines represent the growth of individual clams in the locality.

Data Exploration:

This section will explore the data, looking to familiarize with the data's distributions and identify initial relationships between clam growth and water body.

Summary Statistics:

Table 2: Summary Statistics of Total Length: Lakes vs. River

body	mean	median	sd	max_growth_period
L	47.3	46.25	12.9	NA
R	62.81	62.52	13.32	39

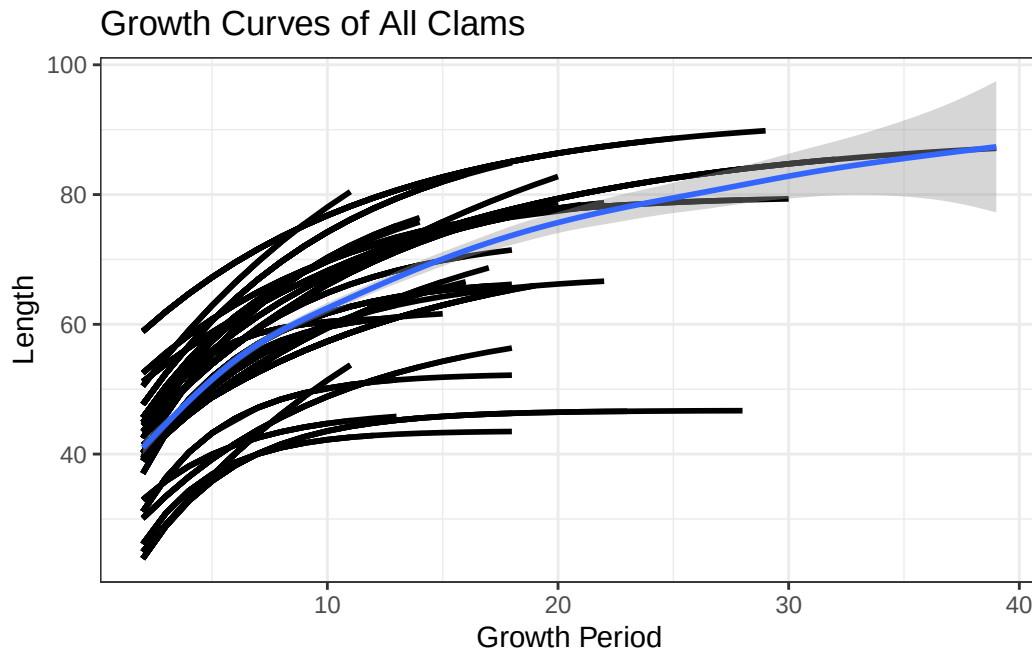
Lakes have smaller average total length then rivers. However, they are both similarly distributed.

Table 3: Summary Statistics of Growth: Lakes vs. Rivers

growth_period	body	mean	median	sd
2	L	6.291	5.82	3.731
2	R	7.995	7.5	3.972
3	L	5.123	4.43	3.007
3	R	5.22	4.95	2.485
4	L	3.802	3.61	1.817
4	R	3.675	3.45	1.802
5	L	3.415	3.18	1.967
5	R	3.056	2.735	1.462
6	L	2.737	2.17	1.649
6	R	2.513	2.265	1.297
7	L	2.16	1.905	1.323
7	R	2.116	1.81	1.222
8	L	1.987	1.63	1.532
8	R	1.869	1.595	1.131
9	L	1.833	1.495	1.199
9	R	1.602	1.335	0.9571
10	L	1.458	1.1	1.091

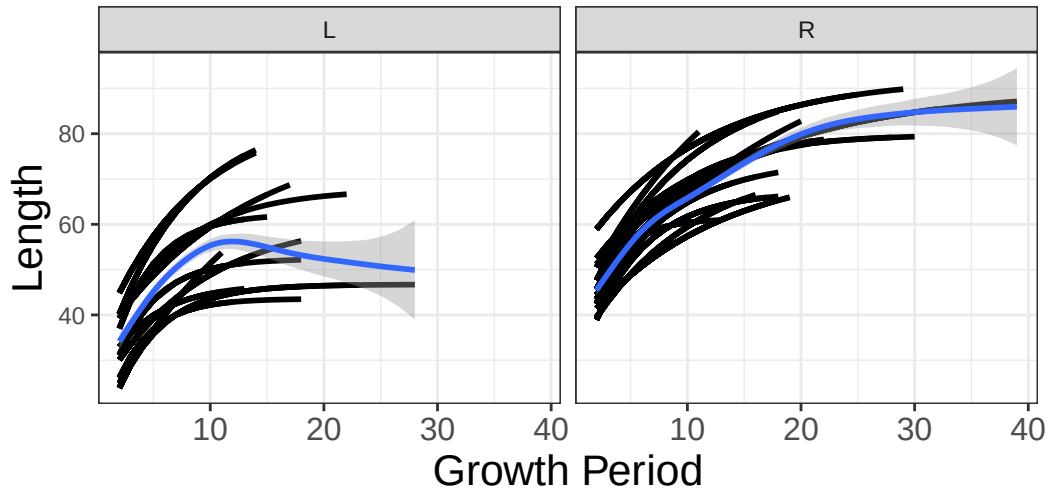
growth_period	body	mean	median	sd
10	R	1.392	1.185	0.7053

Lake and River clams annual growths appear to slow at similar rates.



Here, we used the Von Bertalanffy Growth model on all the clams in the data set. Here we see that the model well illustrates the growth rates of clams. Clams tend to have faster growth rates at the beginning of their life, and slows as they get older.

Growth Curves by Lakes and Rivers



The blue line represents the general Von Bertalanffy growth model for river clams. Black lines represent the growth of each clam in the local

The growth curves for lakes vs. clams vary greatly. They have very different asymptotic lengths. This is because lake clams are much shorter than river clams. Additionally, it appears that lake clams start to reach their asymptotic length quicker than river clams.

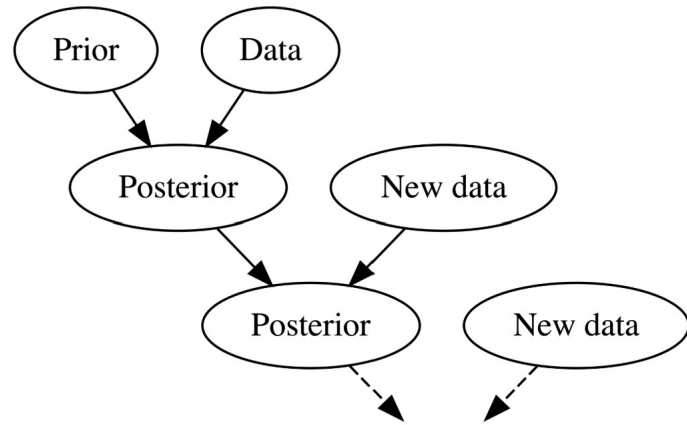
Overall, as we move into our analysis, we can see that growth rate is affected by age making this data a good candidate for the Von Bertalanffy Growth model, but more interestingly the body of water that the clams originate from appears to be an important factor in its growth, especially in terms of overall length achieved and speed at which clams reach this length.

Introduction to Bayesian Statistics:

Fossils, since they are heavily exposed to the outdoors, are often broken or incomplete due to abrasion, erosion, or dissolution. As expected, many of the bivalves in the data are incom-

plete or broken, making it difficult to determine the true start of growth. Therefore, growth periods marked as growth period 1 are not always their first year of life. To account for these uncertainties, a Bayesian statistical framework is used. Bayesian statistics offer a more flexible approach that can account for uncertainty and missing information. This makes it good for this sort of paleontological data, where preservation is imperfect.

Bayesian statistics works by combining prior knowledge with current knowledge to produce an updated understanding, known as the posterior distribution. As additional growth data is considered, the model is continuously updated, improving over time.



Bayesian Algorithm:

For this study, I built an algorithm designed to estimate the parameters of the von Bertanfly growth model while using a Bayesian framework to account for uncertainty in the data. In this algorithm, we want to estimate four parameters of the Von Bertanfly Growth model: $\max_l$, k , b , and σ^2 . The parameter $\max_l$ represents the maximum possible length for clams, k represents the growth rate, b represents time where length is 0, and σ^2

represents the variance. The algorithm runs twice, once for river clams and once for lake clams in order to compare the estimate parameters and results between the two bodies of water.

To begin, we have some data preparation where we clean the data, including data entry errors.

Then, we create and define the von Bertanfly growth function to show shell length as a function of time using asymptotic length (max_l), growth rate (k), and time when length is zero (b).

A key component of this algorithm is the prior and posterior distribution functions, which are computed next. The function `compote_prior_log()` specifies our prior knowledge about the model. It is a probability distribution, in this case a normal distribution, that represents our initials understandings of the data.

The posterior distribution function, `compute_posterior_log`, is the probability distribution function after taking in the new evidence, also defined as a normal distribution.

In simple terms, this part of the code repeatedly tests many possible values for each parameter until it finds values that best fit the data. It begins by setting initial starting values for all parameters. Then, a loop runs 10,000 times, and during each iteration the parameters are updated one at a time. Going parameter by parameter, the loop makes some adjustments, such as increasing or decreasing the parameter, and evaluates how that new value performs. If it improves the model, the new value for the parameter is accepted and held.

The algorithm does this for every parameter until it finds the best combination of parameters for the data. Early on in the iterations, the estimations vary greatly, but as the iterations near

10,000, the model typically converges to one singular value or estimation for the parameter.

```
# get and read data

clams <- read_csv("data/clams_clean.csv")

covariate_data <- read_excel("data/covariate_data.xlsx")

clams <- clams |> left_join(covariate_data, by = "location")


# remove negative values from Coopers Falls, Lake Clear, Mirror Lake

clams <- clams |>

  filter(growth > 0) |>

  mutate(x = case_when(

    body == "L" ~ 1,

    body == "R" ~ 0))


# vb function

vb_function <- function(t, max_l, k, b) {

  max_l * (1 - exp(-k * (t - b)))}


# data for rivers

rivers <- clams |>

  filter(body == "R") |>
```

```

group_by(name)

# MCMC Algorithm

compute_prior_log <- function(max_l, k, b, sigma2) {

  dnorm(max_l, mean = 0, sd = 1000, log = TRUE) +

  dnorm(k, mean = 0, sd = 1000, log = TRUE) +

  dnorm(b, mean = 0, sd = 1000, log = TRUE) +

  log(1 / sigma2)
}

compute_likelihood_log <- function(max_l, k, b, sigma2) {

  dnorm(rivers$Length,

        mean = vb_function(rivers$growth_period, max_l, k, b),

        sd = sqrt(sigma2), log = TRUE) |>

  sum()
}

compute_posterior_log <- function(max_l, k, b, sigma2) {

  compute_prior_log(max_l, k, b, sigma2) +

  compute_likelihood_log(max_l, k, b, sigma2)
}

```

```

}

## set-up the chain

niter <- 10000

# set up three empty vectors, one for each parameter
max_l_store <- rep(NA, 10000)
b_store <- rep(NA, 10000)
k_store <- rep(NA, 10000)
sigma2_store <- rep(NA, 10000)

# define a starting value
max_l_store[1] <- 1.2
k_store[1] <- 0.3183
b_store[1] <- -0.8749
sigma2_store[1] <- 30

for (i in 2:niter) {

```

```

    ## start with max_l

max_l_current <- max_l_store[i - 1]

proposal_sd <- 3

max_l_proposal <- rnorm(1, mean = max_l_current, sd = proposal_sd)

alpha_l <- min(1, exp(compute_posterior_log(max_l_proposal, k_store[i-1], b_store[i-1], si

max_l_store[i] <- sample(c(max_l_proposal, max_l_current), size = 1,
                        prob = c(alpha_l, 1 - alpha_l))

## start K

k_current <- k_store[i - 1]

proposal_sd <- 0.05

k_proposal <- rnorm(1, mean = k_current, sd = proposal_sd)

```

```

alpha_k <- min(1, exp(compute_posterior_log(max_l_store[i], k_proposal, b_store[i-1], sigma2

k_store[i] <- sample(c(k_proposal, k_current), size = 1,

                      prob = c(alpha_k, 1 - alpha_k))

# start B

b_current <- b_store[i - 1]

proposal_sd <- 0.1

b_proposal <- rnorm(1, mean = b_current, sd = proposal_sd)

alpha_b <- min(1, exp(compute_posterior_log(max_l_store[i], k_store[i], b_proposal, sigma2

b_store[i] <- sample(c(b_proposal, b_current), size = 1,

                      prob = c(alpha_b, 1 - alpha_b))

# start sigma2

```

```

sigma2_current <- sigma2_store[i-1]

sigma2_current_log <- sigma2_current |> log()

proposal_sd <- 0.2

sigma2_proposal_log <- rnorm(1, mean = sigma2_current_log, sd = proposal_sd)
sigma2_proposal <- exp(sigma2_proposal_log)

alpha_sigma2 <- min(1, exp(compute_posterior_log(max_l_store[i], k_store[i], b_store[i], s

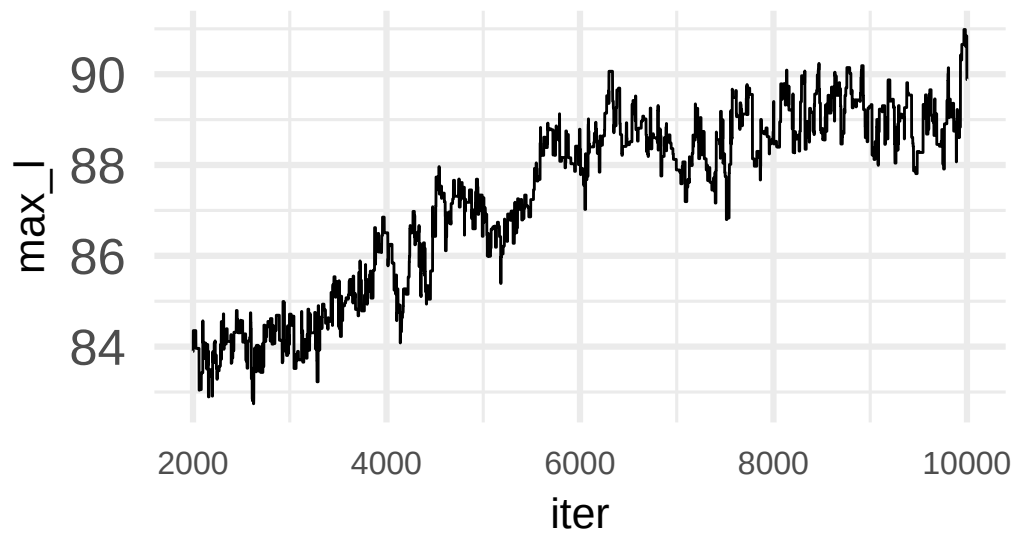
sigma2_store[i] <- sample(c(sigma2_proposal, sigma2_current), size = 1,
                          prob = c(alpha_sigma2, 1 - alpha_sigma2))
}

plot_df_river <- tibble(iter = 1:niter, max_l = max_l_store, k = k_store, b = b_store,
                        sigma2_store) |> filter(iter > 2000)

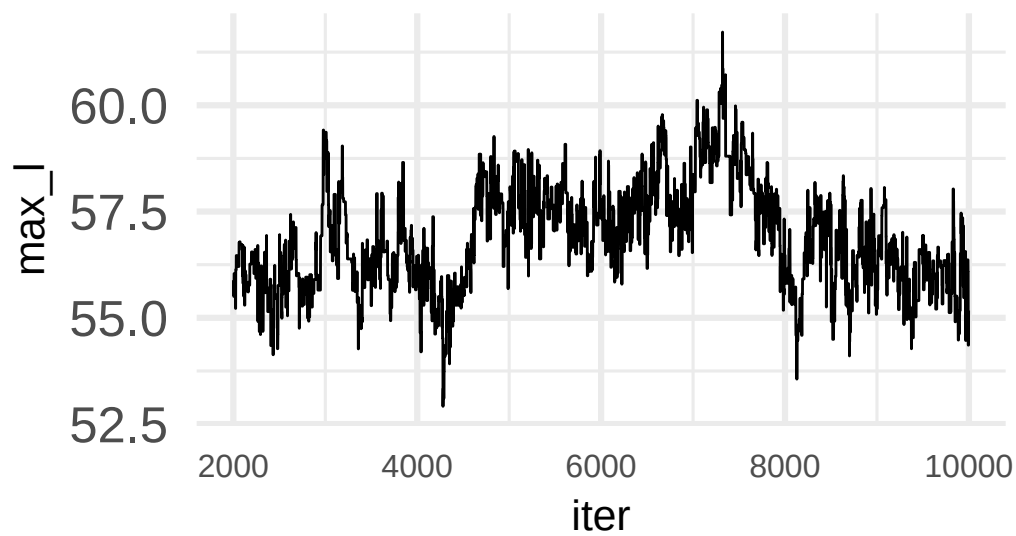
```

Results:

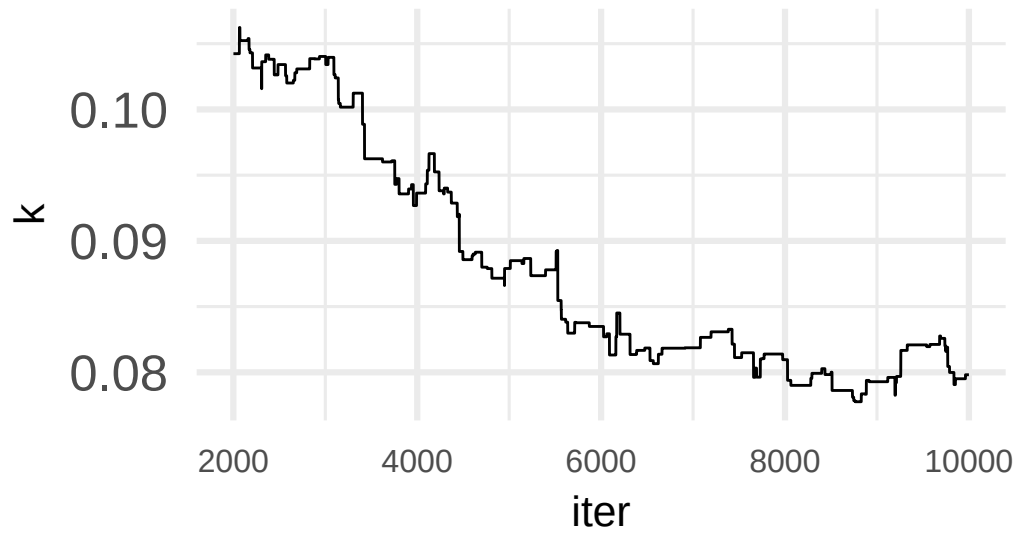
max_l by Iteration for River Clams



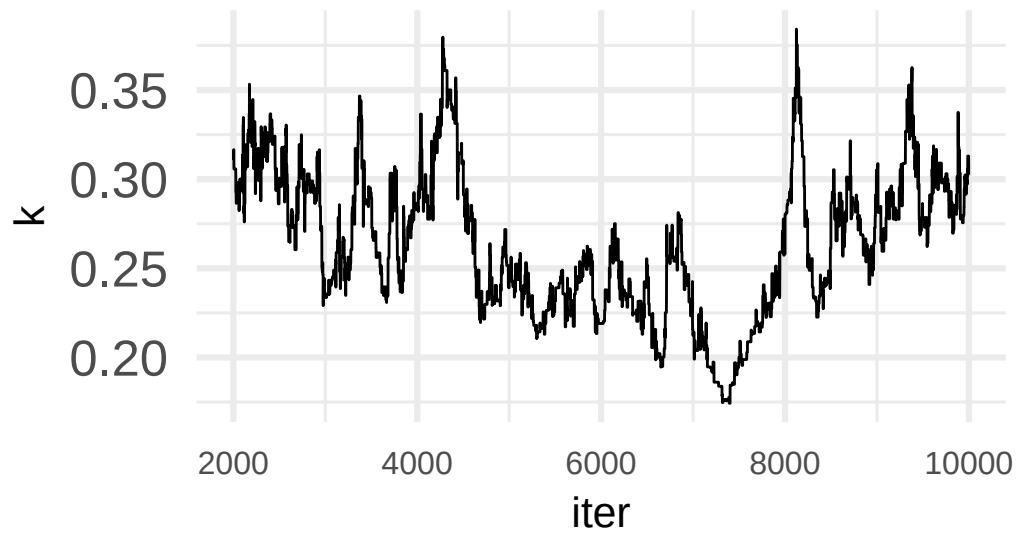
max_l by Iteration for Lake Clams



K by Iteration for River Clams

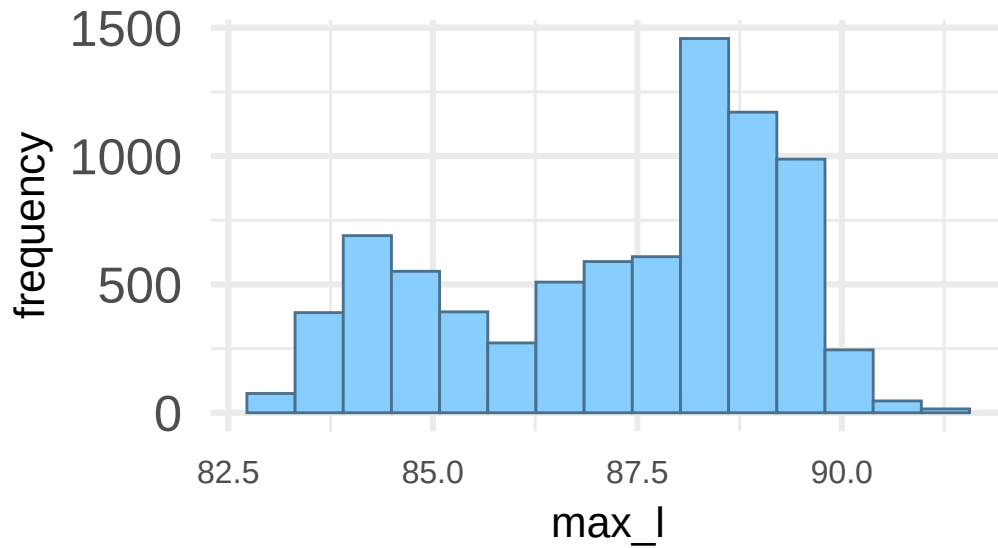


K by Iteration for Lake Clams

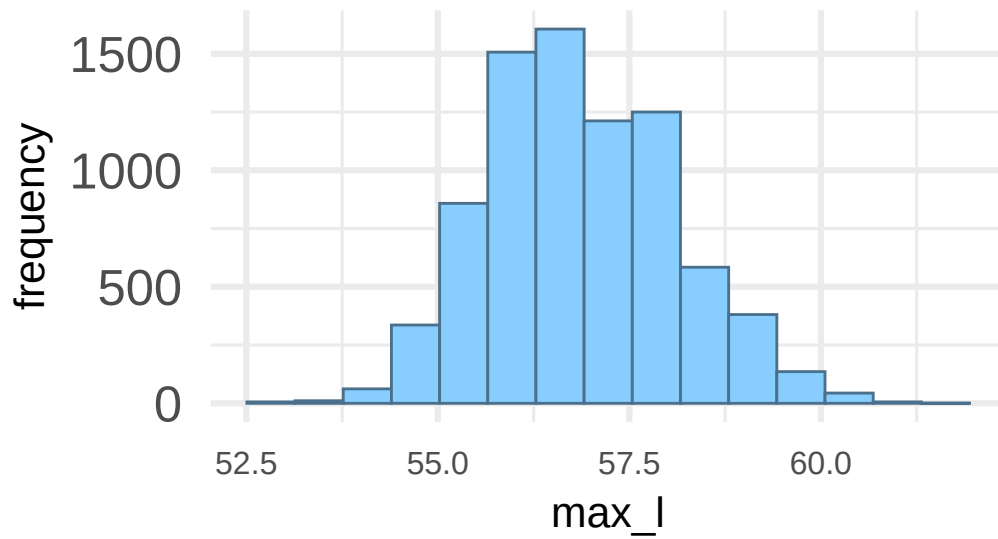


The parameters typically converge for both river and lake clams.

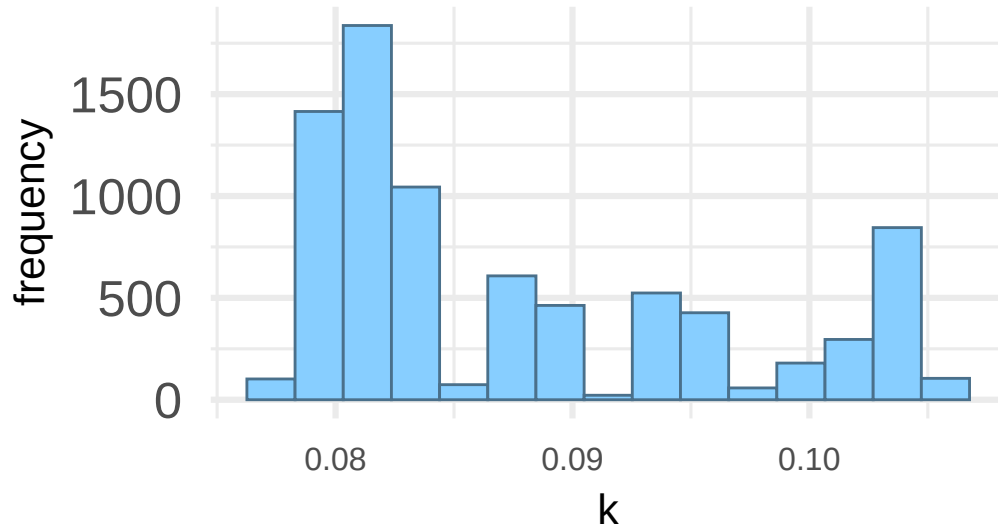
Histogram of max_I for River Clams



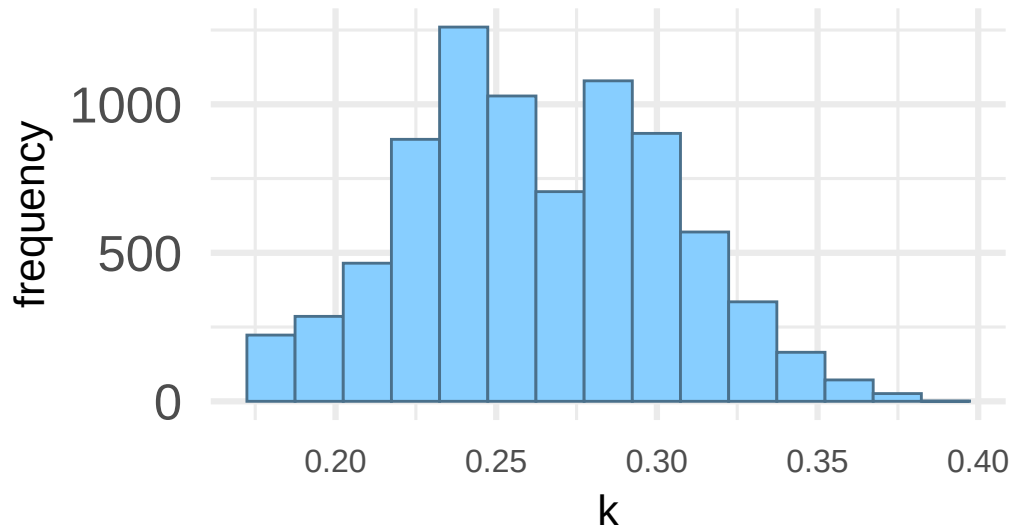
Histogram of max_I for Lake Clams



Histogram of k for River Clams



Histogram of k for Lake Clams



The parameters tested in the algorithm are relatively normal for river and lake clams.

Table 4: Parameter Estimates for Lakes and Rivers

median_l	median_k	median_b	median_sigma2	body
56.72	0.2594	-1.314	114.1	Lake
87.97	0.08328	-6.696	73.34	River

River clams are longer overall in comparison to lake clams. Lake clams stop growing sooner than river clams.

Conclusions and Discussion:

River clams grow to longer overall lengths than lake clams, suggesting that environmental conditions in river aquatic systems that are more favorable for long term growth than lake environments. This could be due to currents, food availability, or acidity. Clam shells are made up of calcium carbonate, which can dissolve in acidic water. Lakes are more acidic than rivers, so one theory suggests that lake clams shells are dissolving as they are growing, explaining the smaller length. Another suggestion is that as filter feeders, clams rely on the water to pass nutrients by them. Since rivers have more aquatic movement, river clams have access to more food, allowing them to grow more.

However, lake clams reach their asymptotic length approximately twice as fast as river clams. This means that they stop growing sooner and reach their maximum length quicker. This suggests that lake environments may support quick early development, but not long-term growth.

Further Study:

There are possible extensions of this study. First, would be to explicitly account for the uncertainty in timing. Currently, the model assumes that the first growth period aligns with the first year of growth. This could be addressed by adding additional parameters into the model and algorithm.

Another extension would be to expand the algorithm so that it estimates parameters for every single location individually. This would allow for study and comparison of individual locations. Using this, you could study which lake or river is the best or worst at supporting clam growth. This approach would look into environmental differences more deeply.

References:

- Johnson, A. A., Ott, M. Q., & Dogucu, M. (2022). Bayes rules! An introduction to applied Bayesian modeling. <https://www.bayesrulesbook.com/>
- Windsland, K., Hvingel, C., Nilssen, E. M., & Sundet, J. H. (2013). Evaluation of von Bertalanffy growth curves for the introduced red king crab (*Paralithodes camtschaticus*) in Norwegian waters. *Fisheries Research*, 145, 15–21. <https://doi.org/10.1016/j.fishres.2013.03.003>
- Thank you to Dr. Judith Nagel-Myers for letting me use her data.